VIKAS SHUKLA

McLean, VA | (716) 903 2501 | *vikasofficial927@gmail.com* | *LinkedIn* | *GitHub* | *Portfolio*

Data Professional with 8+ years of experience building scalable, high-performance systems. Proven technical leader with strong track record of delivering innovative solutions and leading cross-functional teams.

EXPERIENCE

Contractor | Capital One | *McLean, VA* ***Oct 2024 - Present***

Cards Platform Engineering & Enterprise Services

- Built automated Python/SQL data quality monitoring system with real-time KPI alerting, catching data issues before business impact in ~80% of cases and reducing weekly metric variance by 20%.
- Consolidated 40M+ transactions across disparate systems into a unified self-serve analytics layer, eliminating 60% of manual analysis cycles.
- Productionized fraud scoring models with automated threshold management via CI/CD pipeline, achieving 98% model adoption across Risk Management teams.
- Partnered with engineering and compliance to align AI model outputs with U.S. regulatory standards; documented diagnostics and root-cause analysis fixes.
- Executed Risk Management and Audit testing using Test of Design and Test of Effectiveness for data pipeline controls. Mapped controls to risks.
- Developed and optimized complex PySpark scripts in Databricks, executing 70+ daily credit card controls to validate 15M+ accounts across APR, fees, payments, and statements.
- Managed campaign runs, production catch-up reruns, and script fixes, resolving data discrepancies, execution failures, and reconciliation mismatches across Cards Platform Engineering.
- Delivered high-level control performance reports, tracking execution status, mismatch trends, and risk indicators while supporting Script Governance, OE/DE testing, and audit readiness.

Sr. Data Analyst Contract | Bayer | *St. Louis, MO* ***July 2023 - Oct 2024***

- Transformed aging analytics environment into clean and modern reporting system. Rebuilt RMarkdown workflows, containerized all analytical steps in Docker, and delivered platform that ran smoothly every time reporting failures dropped by 90%.
- Implemented scheduled job automation to generate updated reports on monthly basis, ensuring stakeholders receive most recent insights regularly.
- Operationalized R Shiny application for breeding data visualization and model training. Added logistic regression with regularization and stratified cross validation, and exposed ROC and PR curves with simple threshold tuning. Improved AUC from 0.78 to 0.86, lifted F1 by 14%, and reduced iteration time from 3 days to 1.
- Produced complex plots and tables for large data analysis in LaTeX reports.

Software Engineer | Amazon | *Bellevue, WA* ***Jun 2022 - Apr 2023***

- Designed and deployed secure machine learning development infrastructure supporting model training, experimentation, and deployment workflows for the Alexa Data Science organization.
- Built AI-powered notification quality assessment framework integrating user settings, reachability signals, and A/B test results across push, SMS, and email channels, improved Alexa user engagement and retention by 20%.
- Launched uplift-based targeting system with guardrails against notification fatigue, reduced total sends by 22% with no engagement loss, cut SMS/email failures by 30%, and lifted incremental conversion by 8%.
- Built scalable REST APIs and backend services following object-oriented design principles, supporting production experimentation platforms and distributed data pipelines.

Data Analyst | SGS Tekniks | *Gurugram, IND* ***Nov 2014 - May 2018***

- Automated on-premises server fleet scaling for SGS tax engine during peak holiday traffic, reducing deployment time from 3-4 hours per server to just 25 seconds.
- Led the infrastructure design and deployment of SGS tax processing platform, ensuring high availability, scalability, and reliable performance during peak sales periods.
- Mentored interns, organized innovation hackathons, and delivered technical training programs to strengthen team capability and engineering excellence.

KEY PROJECTS

Fraud Detection Project | *Stack: Python, SQL* GitHub

- Engineered features (target encoding, transaction patterns), cleaned redundant attributes, and trained XGBoost classifier.
- Validated performance using decile analysis, Lorenz curve and Gini Coefficient for discrimination strength.
- Top two decile captures majority of fraudulent transaction, achieved high recall while minimizing false positives.

Forecast Bike Rentals | *Stack: Python, SQL* GitHub

- Identified and plotted features gave high and low correlation with response variable. And anticipate total rentals per hour for last quarter of 2012 by using gradient boosting algorithm.
- Estimated total rentals per hour for last quarter of 2012.

IntelliQuery Engine | *Stack: Python, SQL* GitHub

- Implemented agentic routing and context-aware retrieval workflows for finance/legal queries, improving accuracy by 45% and reducing manual validation effort by 50%.
- Deployed scalable FastAPI backend integrated with React Native application delivering production-grade low-latency AI responses.

EDUCATION

M.S. Data Science (GPA: 3.7 / 4.0) | University at Buffalo ***Feb 2022***
B.Tech. Electrical & Electronics Engineering (GPA: 8.6 / 10) | SRM University, India ***May 2014***

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, PySpark, Java
Engineering: Object-Oriented Programming, Scalable Architecture, Git
Cloud & DevOps: AWS Services (Lambda, S3, ECS Fargate, DynamoDB, RDS, SQS, SNS, Kinesis, CloudFormation, Glue, EventBridge, AppConfig), Databricks, QuickSight, Snowflake, Excel